

Lab 3 Grading Rubric Description

Lab 3 has a total of **75 points** divided into the following categories:

1. Code style, structure, reproducibility, and data processing - 12 pts
2. Lab 3.1 embeddings and modeling (BoW, Word2Vec, Glove) - 21 pts
3. Lab 3.2 embeddings and modeling (BERT pretrained + fine tuned) - 20 pts
4. Lab 3.3 interpretability and report - 22 pts

1. Code style, structure, reproducibility, and data processing (12 pts)

- **Skeleton/Structure (5 pts)**
 - One point each for having the following files.
 - `run.sh`
 - `environment.yaml`
 - `lab3.tex` or `lab3.ipynb`
 - `lab3.pdf`
 - Note about where pretrained models are stored (huggingface, bridges, SCF)
 - Subtract one point if the submission meets any of these criteria. Total points for Skeleton/Structure cannot go below 0.
 - Report goes over the page limit.
 - Code is included in the report.
 - Data files are included in Git repository.
 - Not clear where pretrained models are stored
- **Code Style (3 pts)**
 - One point each for:
 - Sufficient comments.
 - Sufficient docstrings.
 - Consistent style used.
- **Reproducibility (2 pts)**
 - One point each for the following:
 - No hard-coded paths outside the `lab3` folder.
 - Clear instructions on how to reproduce results.
- **Data Preprocessing (2 pts)**
 - Trimming (1 pt)
 - Make delayed (1 pt)

2. Lab 3.1 embeddings and modeling (BoW, Word2Vec, Glove) - 21 pts

- **3.1 Generate Embeddings (9 pts)**
 - Bag-of-Words (2 pts)
 - Word-to-Vec (2 pts)
 - Glove (2 pts)
 - Description of each model (3 pts)
- **3.1 Modeling and Evaluation (12 pts)**
 - Ridge Regression (3 pts)
 - Cross-validation analysis (3 pts)
 - Detailed analysis of best performing model (3 pts)
 - Detailed analysis of performance over voxels (3 pts)

3. Lab 3.2 embeddings and modeling (BERT pretrained + fine tuned) - 20 pts

- **3.2 BERT (12 pts)**
 - Pre-trained (2 pts)
 - Fine-tuned (4 pts)
 - LORA (2 pts)
 - Performance comparison (2 pts)
 - Description of each model (2 pts)
- **3.2 Modeling and Evaluation for BERT (8 pts)**
 - Ridge Regression (2 pts)
 - Cross-validation analysis (2 pts)
 - Detailed analysis of best performing model (2 pts)
 - Detailed analysis of performance over voxels (2 pts)

4. Lab 3.3 interpretability and report - 22 pts

- **3.3 Interpretability (12 pts)**
 - 2 x story analyzed each with (6 pts x2)
 - SHAP (2 pts)
 - LIME (2 pts)
 - Comparison (2 pts)
- **Stability Check 3.1-3.2 (5 pts)**
 - Relevance of stability check (0-5) 1 pt for each model
- **Writing (5 pts)**
 - Report readability and structure (0-3)
 - Overall quality of figures (0-2)